

DANGOTE CEMENT PLC

Digital KPI Automation & Data Quality System

Technical Appendix

Challenge 4 — Dangote Industries Innovation Challenge

July 2026

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1. System Architecture Overview

1.1 Problem Context

Dangote Cement Plc (DCP) operates four cement plants across Nigeria: Obajana (16.25 Mta, 5 kiln lines), Ibese (12 Mta, 4 lines), Gboko (4 Mta), and Okpella (3 Mta). Plant-performance reporting has depended on manual daily, monthly, and year-to-date KPI data compiled in local Excel files and emailed to management. This process creates six systemic pain points: delays (management decisions on stale data), inconsistent templates across plants, missing KPI records with no enforcement, weak traceability of submissions, no visibility into exceptions until month-end, and manual errors entering reports uncorrected.

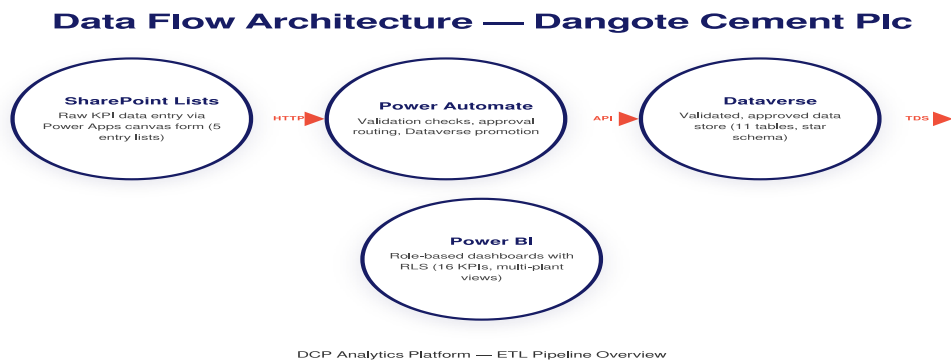
1.2 Solution Architecture

The proposed solution is a full Microsoft Power Platform stack: Power Apps (Canvas) for data entry, SharePoint Lists for entry-tier storage, Power Automate for validation/approval/escalation workflow, Dataverse for validated storage, and Power BI with Row-Level Security for role-based reporting. The architecture follows a two-tier storage pattern to minimise licensing cost while maintaining data integrity.

1.3 Data Flow

The data flow is linear: Plant operators submit raw readings via a standardised Power Apps form backed by SharePoint Lists. Power Automate triggers on new submissions, runs completeness and range validation, and routes clean submissions to the Plant Manager for approval. Approved data is promoted to Dataverse. Power BI reads exclusively from Dataverse, with RLS filtering by the logged-

in Microsoft 365 user's role



1.4 Technology Stack

Layer	Tool	Purpose
Data Entry & Forms	Power Apps (Canvas App)	Standardised KPI input interface for plant operators
Workflow & Validation	Power Automate	Automated validation, sign-off routing, escalation
Entry-Tier Storage	SharePoint Lists	Low-cost raw data capture (standard M365 license)
Validated Storage	Microsoft Dataverse	Clean, relational store with enforced integrity & RLS
Reporting & Dashboards	Power BI	Role-based dashboards with RLS filtering
Identity & Access	Microsoft 365 + Power BI RLS	Authentication via M365; authorisation via data model

1.5 Two-Tier Storage Rationale

Dataverse alone would be the textbook answer but requires Power Apps Premium licensing for every user — expensive across 4 plants with dozens of operators. SharePoint alone is cheap but has no enforced relationships, 5,000-item view limits, and weak integrity. The two-tier design lets operators work in the cheap tier (SharePoint via standard M365 licenses) and only promotes validated, approved data to the expensive, secure tier (Dataverse). Power BI reads only from Dataverse, so what reaches the dashboard is clean by construction.

2. Full KPI Taxonomy

2.1 Core KPIs — Daily (15)

These 15 KPIs are operationalised in the live system. Reported daily, tracked in the Power Apps form, validated by Power Automate, and visible in Power BI.

#	KPI Name	Category	Unit	Target / Benchmark
1	Clinker Production Volume	Production	Tonnes/day	Per plant capacity
2	Cement Production Volume	Production	Tonnes/day	Per plant capacity
3	Kiln Utilization Rate	Production	%	≥ 85%
4	Overall Equipment Effectiveness (OEE)	Production	%	≥ 85%
5	Clinker Factor	Production	Ratio	≤ 0.78
6	Specific Heat Consumption	Energy	kcal/kg clinker	≤ 750 kcal/kg
7	Specific Power Consumption	Energy	kWh/tonne	Per subsystem benchmark
8	Thermal Substitution Rate (TSR)	Energy	%	≥ 10%
9	Free Lime Content	Quality	%	≤ 1.5%
10	Blaine Fineness	Quality	cm ² /g	Per grade specification
11	Off-spec Batch Rate	Quality	%	≤ 2%
12	PM Compliance Rate	Maintenance	%	≥ 90%
13	Mean Time Between Failures (MTBF)	Maintenance	Hours	Trending upward
14	Maintenance Cost per Tonne	Maintenance	₹/tonne	Per budget
15	Submission Timeliness	Governance	Status	100% on-time

Note: KPIs are not entered directly by operators. Operators enter raw input values (meter readings, fuel quantities, downtime hours, lab results). The system calculates KPI values from those inputs. Validation rules apply at both raw input and calculated KPI levels.

2.2 Supporting KPIs — Appendix Layer (~10)

These KPIs are reported weekly or monthly and documented here as part of the full taxonomy but are not operationalised in the daily submission form.

KPI	Category	Frequency
Lime Saturation Factor (LSF)	Quality	Daily (lab)
Silica Modulus (SM)	Quality	Daily (lab)
Alumina Modulus (AM)	Quality	Daily (lab)
28-day Compressive Strength	Quality	Weekly
Mean Time to Repair (MTTR)	Maintenance	Monthly
Reactive vs. Planned Maintenance Ratio	Maintenance	Monthly
Fuel Cost per Tonne of Clinker	Energy/Finance	Monthly
TSR Detail by Fuel Type	Energy	Monthly
Packing & Dispatch Volume	Production	Daily
Kiln Feed Rate	Production	Daily (SCADA)
Plant Capacity Utilization Rate	Production	Daily (calculated)
Manufacturing Cost per Tonne	Finance	Monthly

3. Data Model Summary

The system uses a star schema of 11 tables — 4 dimension tables holding reference masters and 7 fact tables holding operational data. All relationships are Many-to-One; there are no Many-to-Many relationships. Full column definitions, data types, and relationships are shown in the Entity Relationship Diagram in Section 4.

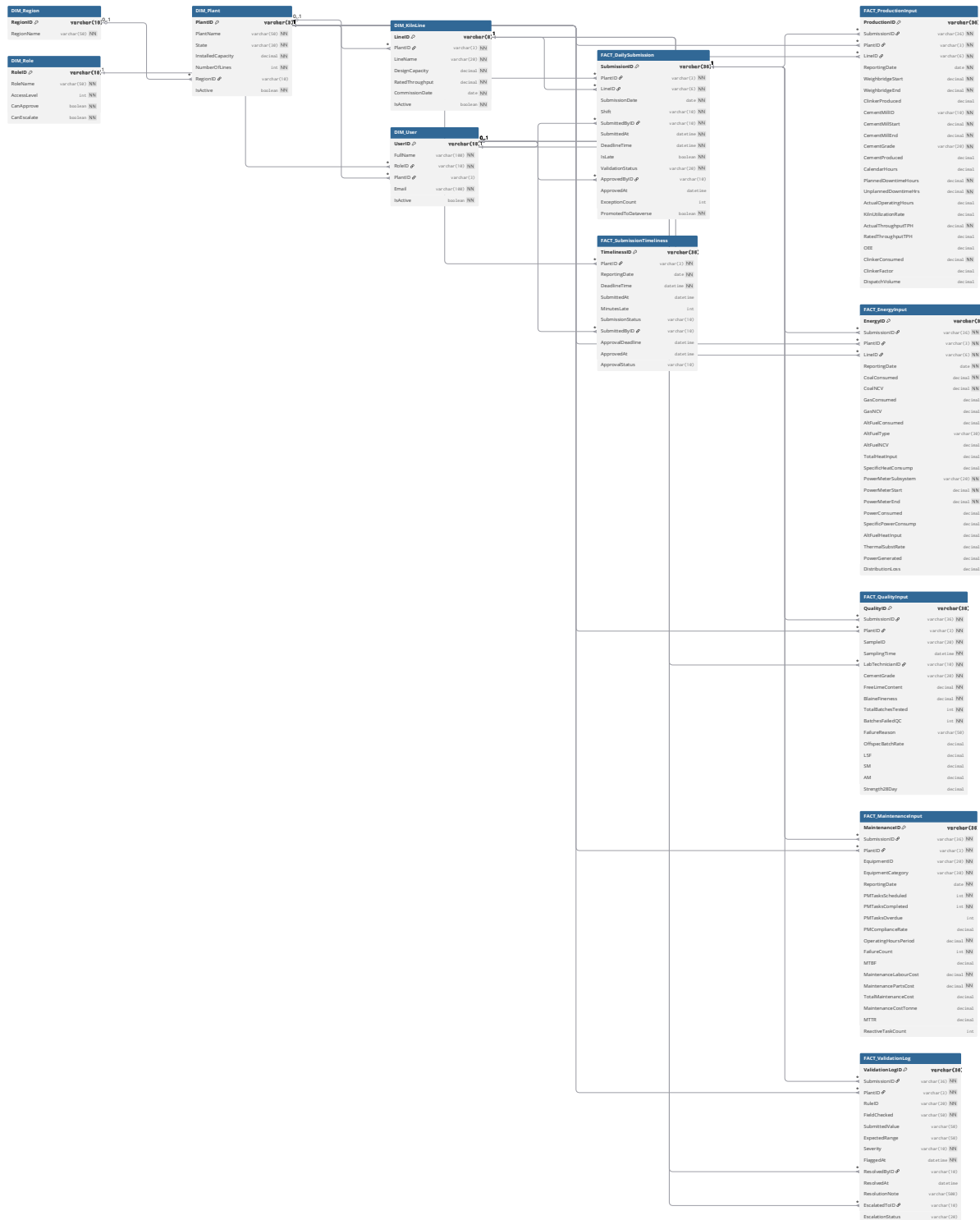
Table	What It Stores	Storage Tier
DIM_Plant	One record per plant — capacity, location, line count	Dataverse
DIM_KilnLine	One record per kiln line — design capacity, rated throughput	Dataverse
DIM_User	One record per system user — role, plant, M365 email (RLS key)	Dataverse
DIM_Role	Five system roles and their approval/escalation permissions	Dataverse
FACT_DailySubmission	Master record per plant per day — tracks submission lifecycle	SharePoint → Dataverse
FACT_ProductionInput	Raw inputs for KPIs 1–6 (clinker, cement, OEE, utilization)	SharePoint → Dataverse
FACT_EnergyInput	Fuel consumption and power readings for KPIs 7–9	SharePoint → Dataverse
FACT_QualityInput	Lab test results and batch QC data for KPIs 10–12	SharePoint → Dataverse
FACT_MaintenanceInput	Equipment maintenance data for KPIs 13–15	SharePoint → Dataverse
FACT_ValidationLog	System-generated exception and audit trail — never user-written	Dataverse only
FACT_SubmissionTimeliness	Auto-generated KPI 16 record per plant per day	Dataverse only

See the Entity Relationship Diagram in Section 4 for the full column-level definition, data types, primary and foreign keys, and inter-table relationships.

4. Entity Relationship Diagram

The Entity Relationship Diagram below shows all 11 tables, their columns, primary and foreign keys, and the relationships between them.

[click here for clearer image](#)



5. SharePoint & Power Apps Configuration Summary

5.1 SharePoint List Structure

Five SharePoint lists form the data entry layer. Each list mirrors a FACT table in the data model (Section 3). Operators enter data via Power Apps, which writes directly to these lists. Power Automate validates the data and promotes clean records to Dataverse for Power BI reporting.

List Name	Maps To	Purpose	Columns
Production Reports (Master)	FACT_DailySubmission	Tracks submission lifecycle, approval, deadlines	14 total (8 visible, 6 system)
Production Inputs	FACT_ProductionInput	Clinker/cement volumes, OEE, utilization (KPIs 1–6)	22 (10 visible, 12 calc/system)
Energy Inputs	FACT_EnergyInput	Fuel consumption, power readings (KPIs 7–9)	17 (8 visible, 9 calc/system)
Quality Inputs	FACT_QualityInput	Lab results, QC batches (KPIs 10–12)	14 (7 visible, 7 calc/system)
Maintenance Inputs	FACT_MaintenanceInput	Equipment PM, failures, costs (KPIs 13–15)	16 (8 visible, 8 calc/system)

DCP_DailySubmission ☆

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All Items

+ Add view

SubmissionID	PlantID	LineID	SubmissionDate ⓘ	Shift	SubmittedByID	DeadlineTime ⓘ	ValidationStatus	IsLate	ExceptionCount	PromotedToData...	Appr...
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Welcome to your new list
Select the New button to get started.

Image of FACT_DailySubmission produced in sharepoint

5.1 Power Apps form design

The KPI entry interface was designed as a Power Apps canvas app with tab-based navigation across four input categories — Production, Energy, Quality, and Maintenance. A context strip pinned below the navigation tabs displays the active Plant, Line, Date, Shift, and Operator at all times. Numeric fields are auto-calculated in real time from operator inputs — Clinker Produced, OEE, Kiln Utilization Rate, and Clinker Factor are derived automatically and displayed as read-only fields. On the Quality tab, a dynamic warning banner activates when Free Lime Content exceeds 1.5%, alerting the operator before submission. On submit, the form writes to all five SharePoint lists simultaneously via the Power Apps Patch() function, triggering the Power Automate validation flow.

The screenshot displays the 'Daily KPI Submission' form in a Power Apps interface. The top navigation bar includes a back button, the title 'Daily KPI Submission', the plant name 'Obajana Plant', the date '07 Jul 2026', the shift 'Day Shift', and a user profile icon 'AO'. Below the navigation bar are four tabs: 'Production' (selected), 'Energy', 'Quality', and 'Maintenance'. A context strip below the tabs shows the following information: PLANT: Obajana, LINE: L1, DATE: 07 Jul 2026, SHIFT: Day, OPERATOR: Adebayo O.

The form is divided into several sections:

- Clinker Weighbridge**: Contains two input fields for 'Weighbridge — Start * (Tonnes)' (18450.0) and 'Weighbridge — End * (Tonnes)' (20980.0). Below these is a read-only field for 'Clinker Produced (T/day)' (2,530.0) with a label 'Auto-calculated'.
- Cement Mill**: Contains a dropdown for 'Mill ID *' (CM-01), a read-only field for 'Counter — Start * (Tonnes)' (12500.0), a read-only field for 'Counter — End * (Tonnes)' (14750.0), a dropdown for 'Cement Grade *' (OPC 42.5), a read-only field for 'Cement Produced (T/day)' (2,250.0) with a label 'Auto-calculated', and a read-only field for 'Clinker Consumed * (Tonnes)' (1750.0).
- Utilization & OEE**: Contains a read-only field for 'Planned Downtime * (Hours)' (2.0), a read-only field for 'Unplanned Downtime * (Hours)' (1.5), a read-only field for 'Actual Operating Hours (Hours)' (20.5) with a label 'Auto-calculated', a read-only field for 'Actual Throughput * (TPH)' (125.4), and a read-only field for 'Rated Throughput (TPH)' (145.0) with a label 'From DIM_KilnLine'.
- Dispatch (optional)**: Contains a read-only field for 'Dispatch Volume (T/day)' (—) with a label 'Bagged + bulk cement dispatched'.

At the bottom of the form, there is a status bar showing '5 validations pending • Auto-saves every 60s' and two buttons: 'Save Draft' and 'Submit'.

Daily KPI Submission

Obajana Plant

07 Jul 2026

Day Shift

AO

Production

Energy

Quality

Maintenance

PLANT

Obajana

LINE

L1

DATE

07 Jul 2026

SHIFT

Day

OPERATOR

Adebayo O.

Fuel — Coal

Coal Consumed * (Tonnes)

320.5

Coal NCV * (kcal/kg)

6250

Heat from Coal (Gcal)

2,003.1

Auto-calculated

Fuel — Gas (conditional)

Gas Consumed (MSCF)

—

Only if gas-fired

Gas NCV (kcal/MSCF)

—

Only if gas-fired

Fuel — Alternative (conditional)

Alt Fuel Consumed (Tonnes)

12.0

Alt Fuel Type

RDF

Alt Fuel NCV (kcal/kg)

3800

Heat from Alt Fuel (Gcal)

45.6

Auto-calculated

Power Consumption

Subsystem *

Kiln

Meter — Start * (kWh)

88450

Meter — End * (kWh)

94280

Power Consumed (kWh)

5,830

Auto-calculated

Power Generated (MWh)

—

Captive power (optional)

6 validations pending • Auto-saves every 60s

Save Draft

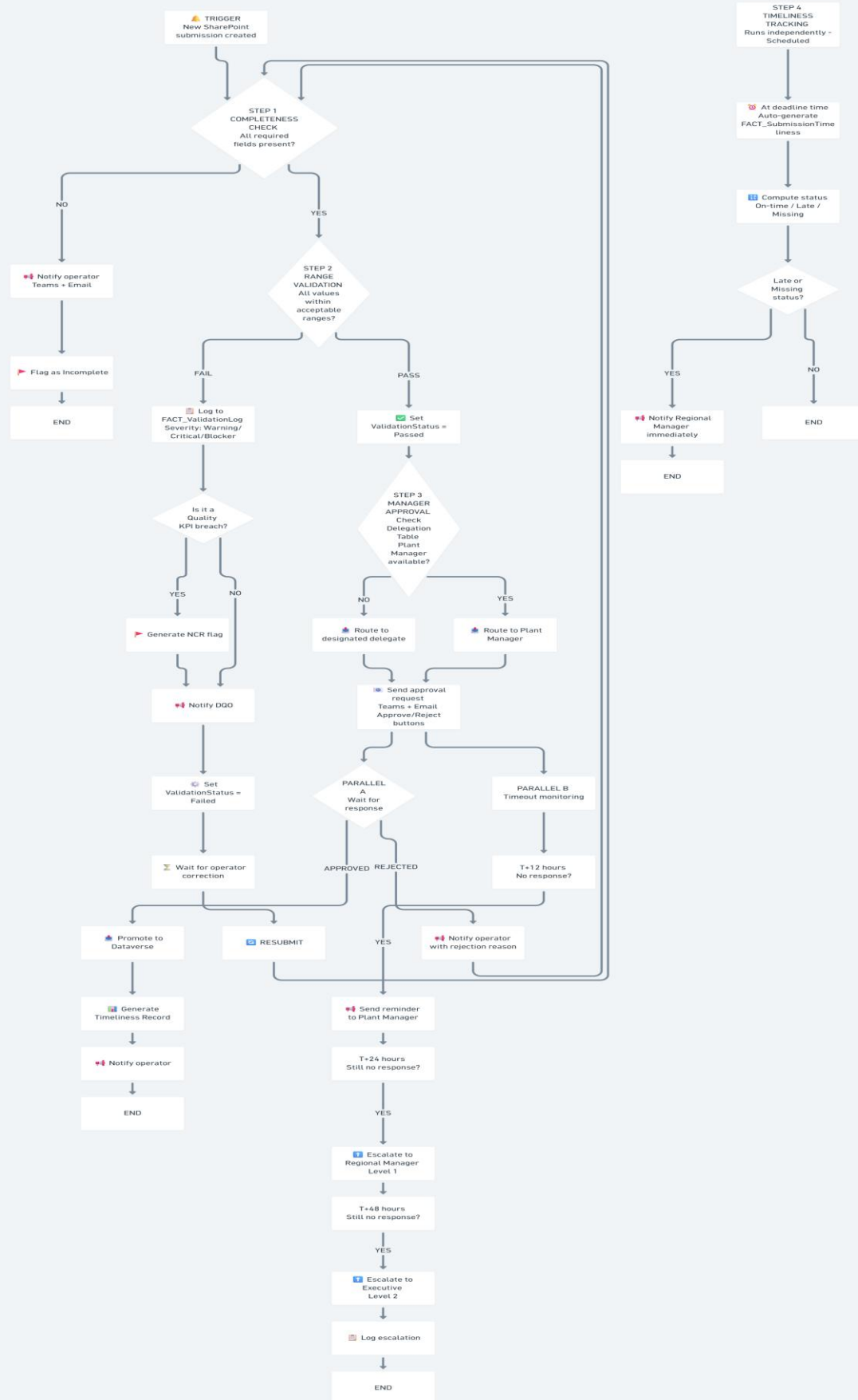
Submit

2 of 4 tabs of the Power Apps Form Page

6. Power Automate Flow Diagram

The Power Automate flow manages the complete submission lifecycle: trigger, validation, sign-off, escalation, and Dataverse promotion.

[click here for a clearer image](#)



6.1 Flow Steps

Trigger: Flow triggers when a new item is created in any of the 5 SharePoint lists.

Completeness Check: Verifies all required fields (marked REQ in the schema) have values. Missing fields are flagged with severity "Blocker".

Range Validation: Validates numeric values are within acceptable bounds. For example: Free Lime Content must be $\leq 1.5\%$; Kiln Utilization Rate cannot exceed 100%; no negative values for any measured quantity.

Decision Branch: If all checks pass: route to Plant Manager for approval. If any check fails: log exception to FACT_ValidationLog, notify Data Quality Officer, and raise an NCR (Non-Conformance Report) flag against the submission.

Delegation Check: If the Plant Manager has set a delegate (approval proxy) for the current period, the approval request is routed to the delegate instead.

Approval Process: A Power Automate Approval is sent to the Plant Manager (or delegate). If approved within SLA: mark PromotedToDataverse = True and copy record to Dataverse.

T+12 Reminder: If no action is taken within 12 hours of submission, an automated reminder is sent to the Plant Manager and the DQO is copied.

First Escalation (T+24): If still not approved after 24 hours: the submission is escalated to the Regional Manager, and the Plant Manager is notified of the escalation.

Second Escalation (T+48): If still not approved after 48 hours: the submission is escalated to the Executive level, and all prior recipients are notified of the escalation chain.

Re-submission Loop: If the submission is rejected at any approval level, it is returned to the operator with rejection notes. The operator corrects it and re-submits, re-entering the validation pipeline from the Completeness Check step.

Timeliness Check: At the configured deadline (configurable per plant), the flow computes whether each submission was on-time, late, or missing. Results are written to FACT_SubmissionTimeliness.

7. Validation Rule Library

The validation rule library is organised into four categories: completeness rules, range rules, consistency rules, and timeliness rules. These rules are implemented in Power Automate and triggered on every new submission.

7.1 Completeness Rules

Rule ID	Field	Rule	Severity	KPI Affected
CMP-01	SubmittedByID	Must not be null	Blocker	All
CMP-02	SubmissionDate	Must not be null	Blocker	All
CMP-03	Shift	Must be Day, Night, or Full Day	Blocker	All
CMP-04	WeighbridgeStart	Must not be null	Blocker	KPI 1
CMP-05	WeighbridgeEnd	Must not be null	Blocker	KPI 1
CMP-06	CementMillID	Must not be null	Blocker	KPI 2
CMP-07	CementMillStart	Must not be null	Blocker	KPI 2
CMP-08	CementMillEnd	Must not be null	Blocker	KPI 2
CMP-09	PlannedDowntimeHours	Must not be null	Warning	KPI 3
CMP-10	UnplannedDowntimeHrs	Must not be null	Warning	KPI 3
CMP-11	CoalConsumed	Must not be null	Blocker	KPI 6
CMP-12	CoalINCV	Must not be null	Blocker	KPI 6
CMP-13	SampleID	Must not be null	Blocker	KPI 9, 10
CMP-14	FreeLimeContent	Must not be null	Blocker	KPI 9
CMP-15	BlaineFineness	Must not be null	Blocker	KPI 10
CMP-16	PMTasksScheduled	Must not be null	Blocker	KPI 12
CMP-17	PMTasksCompleted	Must not be null	Blocker	KPI 12
CMP-18	FailureCount	Must not be null	Blocker	KPI 13
CMP-19	MaintenanceLabourCost	Must not be null	Warning	KPI 14
CMP-20	MaintenancePartsCost	Must not be null	Warning	KPI 14

7.2 Range Rules

Rule ID	Field	Rule	Severity	KPI Affected
RNG-01	ClinkerProduced	Must be > 0 and <= plant daily capacity	Critical	KPI 1
RNG-02	CementProduced	Must be > 0 and <= plant daily capacity	Critical	KPI 2
RNG-03	KilnUtilizationRate	Must be 0% to 100%	Critical	KPI 3
RNG-04	OEE	Must be 0% to 100%	Critical	KPI 4
RNG-05	ClinkerFactor	Must be 0.5 to 0.95	Critical	KPI 5
RNG-06	SpecificHeatConsump	Must be 600 to 900 kcal/kg clinker	Critical	KPI 6

RNG-07	FreeLimeContent	Must be <= 1.5%	Critical	KPI 9
RNG-08	OffspecBatchRate	Must be <= 2%	Warning	KPI 11
RNG-09	PMComplianceRate	Must be >= 50% (alert if < 90%)	Warning	KPI 12
RNG-10	WeighbridgeEnd	Must be > WeighbridgeStart	Blocker	KPI 1
RNG-11	CementMillEnd	Must be > CementMillStart	Blocker	KPI 2
RNG-12	PowerMeterEnd	Must be >= PowerMeterStart	Blocker	KPI 7
RNG-13	PlannedDowntimeHours	Must be >= 0 and <= 24	Warning	KPI 3
RNG-14	UnplannedDowntimeHrs	Must be >= 0 and <= 24	Warning	KPI 3
RNG-15	TotalHeatInput	Must be > 0 if clinker produced > 0	Critical	KPI 6

7.3 Consistency Rules

Rule ID	Rule	Severity	Fields Involved	Description
CNS-01	ActualOpHours + Downtime = CalendarHours	Critical	ActualOperatingHours, PlannedDowntimeHours, UnplannedDowntimeHrs, CalendarHours	Operating hours plus all downtime should equal 24 hours for a daily submission
CNS-02	If ClinkerProduced > 0 then CoalConsumed > 0	Warning	ClinkerProduced, CoalConsumed	Cannot produce clinker without fuel input (except during all-electric operation)
CNS-03	PMTasksCompleted <= PMTasksScheduled	Blocker	PMTasksCompleted, PMTasksScheduled	Cannot complete more tasks than were scheduled
CNS-04	BatchesFailedQC <= TotalBatchesTested	Blocker	BatchesFailedQC, TotalBatchesTested	Failure count cannot exceed total test count
CNS-05	ApprovedAt >= SubmittedAt	Blocker	ApprovedAt, SubmittedAt	Approval timestamp cannot precede submission timestamp
CNS-06	PMComplianceRate = (Completed / Scheduled) * 100	Warning	PMComplianceRate, PMTasksCompleted, PMTasksScheduled	Value must match the KPI calculation exactly

7.4 Timeliness Rules

Rule ID	Rule	Severity	Description
TML-01	If SubmittedAt > DeadlineTime then IsLate = True	Critical	Late flag is set automatically by Power Automate
TML-02	If submission missing by DeadlineTime + 1hr, create	Warning	Generates a FACT_SubmissionTimeliness record with status

	"Missing" record		"Missing"
TML-03	T+12: Send reminder if not approved within 12 hours of submission; T+24: Escalate to Regional Manager if still not approved	Warning	Two-stage escalation for management-level SLA breach. Reminder at T+12 copies DQO; escalation at T+24 notifies Regional Manager and Plant Manager.
TML-04	T+48: Escalate to Executive if still not approved after 48 hours	Critical	Final escalation level for critical SLA breaches. All prior recipients are notified of the escalation chain.

8. Power BI Role-Level Security Design

Power BI Row-Level Security (RLS) is implemented using a single report published to the Power BI Service. Five roles are defined in Power BI Desktop, matching the DIM_Role table. The RLS filter is applied through the DIM_User table using the logged-in Microsoft 365 user's email address.

8.1 Roles Defined

Role	Filter Type	Scope
Plant Operator	Row-level (plant-scoped)	Own shift, own line only
Plant Manager	Row-level (plant-scoped)	Full plant, all shifts
Data Quality Officer	None (all data)	All plants, exceptions view
Regional Manager	None (all data)	All 4 plants, operational view
Executive	Column-level (strategic KPIs only)	Strategic and financial KPIs only

DAX filter expression applied to Plant-scoped roles:

```
DIM_User[Email] = USERPRINCIPALNAME()  
&& RELATED(DIM_User[PlantID]) = FACT_DailySubmission[PlantID]
```

8.2 How RLS Works

- User logs into Power BI Service with their Microsoft 365 credentials.
- Power BI checks the user's email against DIM_User.Email.
- The matching DIM_User record determines RoleID and PlantID.
- The DAX filter associated with that role is applied to all visuals.
- Data outside the role's scope is invisible - not just hidden, but filtered at the query level.

8.3 RLS Implementation Steps in Power BI Desktop

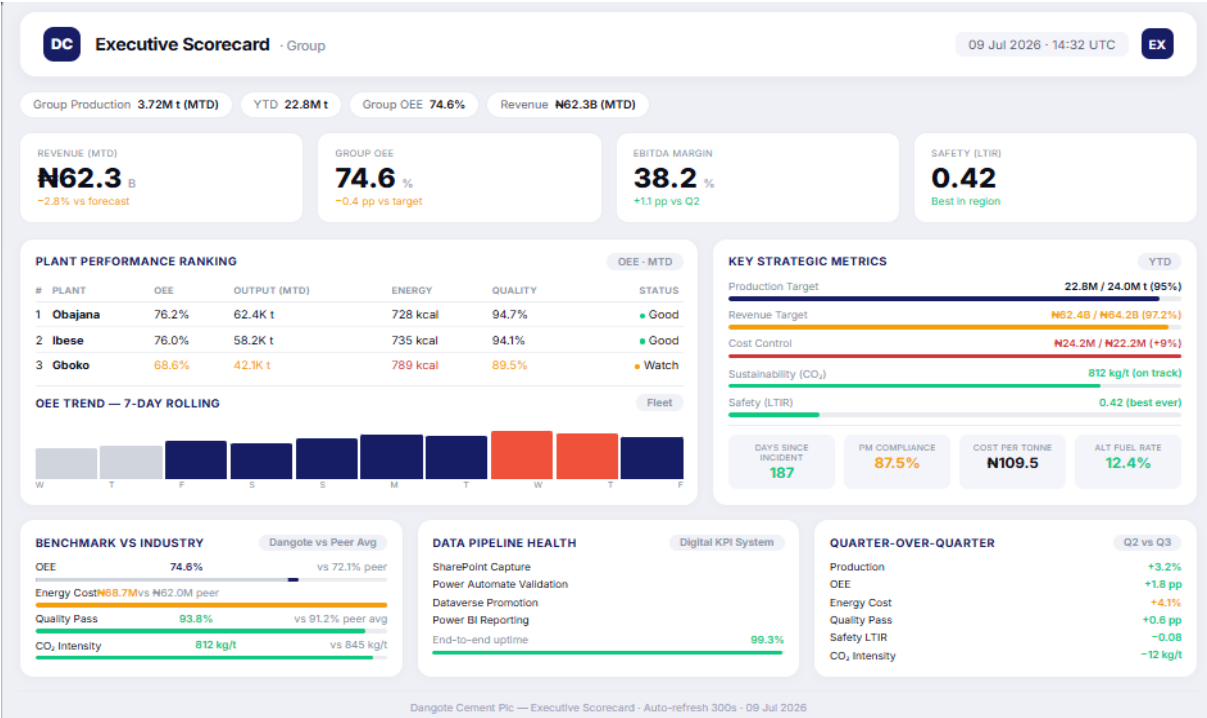
- Create roles in Power BI Desktop (Manage Roles) matching DIM_Role.RoleName (Plant Operator, Plant Manager, Data Quality Officer, Regional Manager, Executive).
- Apply DAX filter: DIM_User[Email] = USERPRINCIPALNAME() for role-based filtering.
- For Plant-scoped roles (Operator, Manager), add: RELATED(DIM_User[PlantID]) = FACT_DailySubmission[PlantID]. This cascades through all relationships.
- Test each role using Power BI Desktop's "View As" function before publishing.
- Publish to Power BI Service and assign members to each role in the Dataset -> Security settings.

8.4 — Power BI Dashboard Design

The Power BI report is structured across four dashboard pages, each filtered by role through Row-Level Security. The Overview page provides a cross-plant KPI scorecard visible to Regional Managers and Executives. The Plant Deep Dive page gives Plant Managers a full view of all 15 operational KPIs for their assigned plant. The Exception Dashboard surfaces all validation flags, severity levels, and resolution status for the Data Quality Officer. The Governance page tracks submission timeliness and approval status across all four plants. All pages are built on the validated Dataverse data store —

ensuring every visual reflects only clean, approved submissions.
All dashboard designs files are [saved here](#)

Executive Scorecard — strategic-level view of group performance (revenue, OEE, EBITDA, safety, production targets across all plants). Designed for board reviews and monthly leadership meetings.



9. Implementation Prerequisites & Assumptions

9.1 Prerequisites

Prerequisite	Details
Microsoft 365 Tenant	Active M365 tenant with SharePoint Online, Power Apps, Power Automate, and Power BI (Pro or Premium per user).
Licensing	Standard M365 E3/E5 for all users (covers SharePoint entry). Power Apps Premium licenses for users accessing Dataverse reports (est. 10-20 users).
Power BI Pro/Premium	Required for RLS functionality and publishing to the service.
SharePoint Lists	5 lists must be created (as defined in Section 5) before the Power Apps form can be connected.
Dataverse Database	Dataverse must be provisioned in the target environment. Default 3GB database capacity should suffice for the pilot.
Plant Champion	One key operator per plant who can act as the power user for training and ongoing support.
M365 Admin Access	Required for DLP policy configuration, SharePoint list creation, user provisioning, and RLS role assignment in Power BI.

9.2 Key Assumptions

- Dangote Cement operates within standard Microsoft 365 infrastructure (Power Platform is deployable without additional data engineering).
- Plant operators have basic digital literacy sufficient for form-based data entry via Power Apps.
- Daily submission deadline is fixed at 10:00 AM following the reporting day (configurable per plant requirement).
- KPI benchmarks and ranges are derived from industry standards and Dangote publicly available documentation.
- All 4 plants and 12 kiln lines are treated as independent reporting units within the same system.
- The Power Automate flow runs in the default Dataverse environment (no custom environment required for pilot).
- Internet connectivity at plant sites is sufficient for Power Apps form submission during scheduled reporting windows.
- Synthetic data used for the prototype is representative of actual plant operations. Real-world patterns may differ.

9.3 Data Limitations

The following data limitations apply to this submission:

- Synthetic data was generated for the prototype and does not represent actual Dangote Cement plant operations.
- KPI targets and benchmarks are based on publicly available industry standards and may differ from DCP's internal targets.
- Validation rule thresholds will require calibration during the pilot phase using actual plant data.

- The data model assumes each submission covers one shift per kiln line. Plants operating on different shift patterns may require schema adjustments.
- Cost estimates for licensing are based on published Microsoft pricing and may differ based on DCP's enterprise agreement.